

Base Case

possible integer n , $n = 5$

LHS

RHS

$$2^5 = 32$$

$$5^2 = 25$$

$$32 > 25 \text{ True}$$

Assume the statement is true

$$k > 4$$

$$2^k > k^2$$

$$2^{k+1} > (k+1)^2$$

$$2^{k+1} = 2 \cdot 2^k$$

$$2^{k+1} = 2k^2$$

$$(k+1)^2 = k^2 + 2k + 1$$

$$2k^2 > k^2 + 2k + 1$$

$$k^2 > 2k + 1$$

$$k^2 - 2k - 1 > 0$$

$\therefore$ True for all $k \geq 3$, since $k > 4$,

The base case is true and inductive

step is valid

$\therefore$ By induction, $2^n > n^2$ for all integers $n > 4$